

1.9 Solving Polynomials Inequalities-Solution

1.Solve the following polynomial inequalities, $x \in \mathbb{R}$.

a) $x^2 - 4x + 3 < 0$

$$(x-3)(x-1) < 0$$

$$x \in (1, 3)$$

b) $x^3 - 3x - 2 \geq 0$

$$(x+1)^2(x-2) \geq 0$$

$$x = -1 \cup [2, \infty)$$

c) $x^4 - 1 \geq 0$

$$(x^2+1)(x-1)(x+1) \geq 0$$

$$x \in (-\infty, -1] \cup [1, \infty)$$

d) $-x^2 + 3x + 1 < 0$

$$-(x^2 - 3x - 1) < 0$$

$$x \in \left(-\infty, \frac{3-\sqrt{13}}{2}\right] \cup \left[\frac{3+\sqrt{13}}{2}, \infty\right)$$

e) $-2x^4 - 2x^3 + 16x^2 + 24x < 0$

$$-2x(x+2)^2(x-3) < 0$$

$$x \in (-\infty, -2) \cup (-2, 0) \cup (3, \infty)$$

f) $2(x+3)(x-1)^2(x-5) \leq 0$

$$x \in [-3, 5]$$

g) $-3(x+4)(x-3)^3 > 0$

$$x \in (-4, 3)$$

h) $x^4 < 22x^2 + 75$

$$x^4 - 22x^2 - 75 < 0$$

$$(x^2 - 25)(x^2 + 3) < 0$$

$$(x-5)(x+5)(x^2+3) < 0$$

$$x \in (-5, 5)$$

i) $2x^2 - 2x \geq 2 - x$

$$2x^2 - x - 2 \geq 0$$

$$x \in \left(-\infty, \frac{1-\sqrt{17}}{4}\right] \cup \left[\frac{1+\sqrt{17}}{4}, \infty\right)$$

2. Let $f(x) = -2x + 1$, $g(x) = x^2 - 2x + 1$ and $h(x) = x^3 - 1$. Determine all values of x such that $f(x) < g(x) < h(x)$ and illustrate the situation graphically.

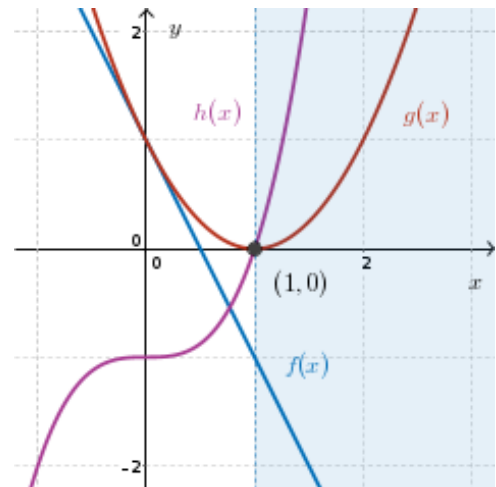
$$-2x + 1 < x^2 - 2x + 1 < x^3 - 1 \Leftrightarrow x^2 - 2x + 1 < x^3 - 1 \rightarrow x^3 - x^2 + 2x - 2 > 0 \quad (1)$$

$$-2x + 1 < x^2 - 2x + 1 \rightarrow x^2 > 0 \quad (2)$$

$$x^3 - x^2 + 2x - 2 > 0 \quad (1)$$

$$\left. \begin{array}{l} (x-1)(x^2+2) > 0 \rightarrow x > 1 \\ x^2 > 0 \quad (2) \rightarrow x \in \mathbb{R}, x \neq 0 \end{array} \right\} \xrightarrow{\cap} x > 1$$

Graphically, this solution is highlighted below:



3. The number n (**in hundreds**), of mosquitoes in a camping area after t weeks can be modelled by the equation $n(t) = 2t^4 - 5t^3 - 16t^2 + 45t$. According to this model, when will the population of mosquitoes be greater than 1800?

$$2t^4 - 5t^3 - 16t^2 + 45t > 18$$

$$2t^4 - 5t^3 - 16t^2 + 45t - 18 > 0$$

$$(t-2)(2t^3 - t^2 - 18t + 9) > 0$$

$$(t-2)(2t-1)(t-3)(t+3) > 0$$

$$t \in \left(\frac{1}{2}, 2\right) \cup (3, \infty)$$

4. A zoo wishes to construct an aquarium in the shape of a rectangular prism such that the length is twice the width and 5 m greater than the height. If the aquarium must have a volume strictly between 1125 m^3 and 3000 m^3 , determine the restrictions on the length of the aquarium.

Let x represent the width of the aquarium ($x > 0$). Then its length is $2x$, its height is $2x - 5$, and the volume of the aquarium is

$$V(x) = x(2x)(2x - 5) = 4x^3 - 10x^2$$

$$1125 < 4x^3 - 10x^2 < 3000$$

or equivalently, $1125 < 4x^3 - 10x^2$ and $4x^3 - 10x^2 < 3000$.

Solving $1125 < 4x^3 - 10x^2$,

$$0 < 4x^3 - 10x^2 - 1125$$

$$0 < (2x - 15)(2x^2 + 10x + 75)$$

Since $(2x^2 + 15x + 150)$ is always positive, the inequality holds when $2x - 15 > 0$, or $x > \frac{15}{2}$

Solving $4x^3 - 10x^2 < 3000$

$$4x^3 - 10x^2 - 3000 < 0$$

$$2(x - 10)(2x^2 + 15x + 150) < 0$$

Since $2(2x^2 + 15x + 150)$ is always positive, the inequality holds when $x - 10 < 0$, or $x < 10$.

Combining these restrictions, the solution to $1125 < V(x) < 3000$ is $\{x \mid \frac{15}{2} < x < 10, x \in \mathbb{R}\}$.

Since the length is $2x$, we see that the length must be between 15 and 20 metres to have the desired volume.

5. Determine the equation of a quintic function $f(x)$ that satisfies the following conditions:

o $f(-3) = f(0) = f(4) = 0$

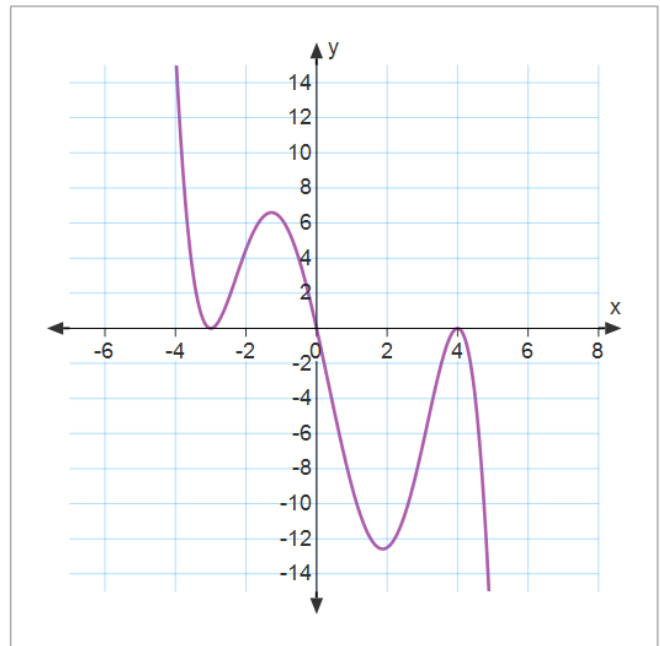
o $f(1) = -9$

o $f(x) > 0$ when $x < -3$ or $-3 < x < 0$

o $f(x) < 0$ when $0 < x < 4$ or $x > 4$

Illustrate the situation graphically.

$$f(x) = -\frac{1}{16}x(x+3)^2(x-4)^2$$



6. The solution to $x^2 + bx + 24 < 0$ is the set of all values of x such that $k < x < k + 2$ for some real value of k . Determine all possible values of b , $b \in \mathbb{R}$. Justify your answer.

Since the solution to $x^2 + bx + 24 < 0$ is the set of all values of x such that $k < x < k + 2$, the expression $x^2 + bx + 24$ must change sign at k and $k + 2$.

Then $k, k + 2$ are the roots of the equation $x^2 + bx + 24 = 0$.

Using the general formula for quadratic equations,

$$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$$

the quadratic with roots k and $k + 2$ is given by

$$x^2 - (2k + 2)x + k(k + 2) = x^2 + bx + 24, \text{ so}$$

$$-(2k + 2) = b \quad \text{and} \quad k(k + 2) = 24$$

$$k^2 + 2k - 24 = 0$$

$$(k + 6)(k - 4) = 0$$

$$k = -6 \text{ or } k = 4$$

When $k = -6, b = 10$.

When $k = 4, b = -10$.

Therefore, $b = -10$ or 10 .

7. A quartic function has turning points at $(-3, 0), (1, 0)$, and $(-1, -16)$. Determine all values of x such that

$$-9 < f(x) < 0$$

$$f(x) = a(x + 3)^2(x - 1)^2 \xrightarrow{(-1, -16)} -16 = a(2)^2(-2)^2$$

$$-16 = 16a$$

$$\therefore f(x) = -(x + 3)^2(x - 1)^2 \quad a = -1$$

$$-9 < f(x) < 0$$

$$-9 < -(x + 3)^2(x - 1)^2 < 0$$

$$\text{or equivalently, } -9 < -(x + 3)^2(x - 1)^2 \text{ and } -(x + 3)^2(x - 1)^2 < 0$$

$$-(x + 3)^2(x - 1)^2 \text{ is always negative when } x \neq 1, x \neq -3$$

$$\text{solving } -9 < -(x + 3)^2(x - 1)^2$$

$$(x + 3)^2(x - 1)^2 - 9 < 0$$

$$[(x + 3)(x - 1) - 3][(x + 3)(x - 1) + 3] < 0$$

$$(x^2 + 2x - 6)(x^2 + 2x) < 0$$

$$x^2 + 2x - 6 = 0 \text{ or } x^2 + 2x = 0$$

$$x = 1 \pm \sqrt{7} \text{ or } x = 0, -2$$

$$\{x \mid -1 - \sqrt{7} < x < -3 \text{ or } -3 < x < -2 \text{ or } 0 < x < 1 \text{ or } 1 < x < 1 + \sqrt{7}, x \in \mathbb{R}\}$$

